

Jharkhand University of Technology, Ranchi
Diploma 5th Semester Examination, 2023

Subject : Java Programming

Subject Code : CSE 504

Time Allowed : 3 Hours

Full Marks : 80

Pass Marks : 26

Answer in your own words.

Answer five questions in which Question No. 1 is compulsory.

All questions carry equal marks.

1. Choose the most suitable answer from the following options:

2×8=16

(i) Which of the following is true about class?

- (a) Class possesses data and methods
- (b) Classes are physical in nature
- (c) Collection of similar type of objects is a class
- (d) Both (a) and (c)

(ii) The concept of derived class is involved in

- (a) Inheritance
- (b) Encapsulation
- (c) Data hiding
- (d) Abstract data types

(iii) What is the name of the tool that is used for compiling a java program?

- (a) Javap
- (b) Java
- (c) Javah
- (d) Javac

(iv) What is the extension of the source files in Java?

- (a) . Jav
- (b) . Java
- (c) . Bytecode
- (d) . Class

(v) What is the output when the following program is compiled and run?

```
Class test {  
    Public static void main (string args[])  
    { int i,j,k, l=0;  
      k = l++;  
      j = ++k;  
      i = j++;  
      System.out.println(i); } }
```

- (a) 0
- (b) 1
- (c) 2
- (d) 3

- (vi) Which is the keyword used for deriving classes?
- (a) Implements (b) Extends
(c) Throws (d) Inherits
- (vii) Which keyword is used for accessing the features of a package?
- (a) Export (b) Import
(c) Package (d) Extends
- (viii) Which tool is used to compile an applet?
- (a) Java (b) Javac
(c) Appletviewer (d) Appletc
2. (a) What are classes and objects? State the different ways of executing on applet.
(b) What are the rules for naming an identifier in Java? Explain the structure of a java program with an example. 8+8
3. (a) Write short notes on polymorphism. Differentiate between Procedural languages and OOPS languages.
(b) Explain the importance of JVM in JRE. Write a program to print "Hello" message in java. 8+8
4. (a) What is a thread? How do threads behave in java?
(b) What are exceptions? How are they handled in java? 8+8
5. (a) Write a program to count the number of words and characters in a string.
(b) Write a program to find the Armstrong Number in Java. 8+8
6. (a) Explain super keyword with all its usages. Support explanation with a program.
(b) What is method overloading? Explain with the help of a program. 8+8
7. Write short notes on any four: 4×4=16
- (a) Dynamic binding
(b) Constructor
(c) Servlet interface
(d) The do statement
(e) Conditional operator
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